Sizing and Synthesis

1 Overview

The definition of aircraft sizing is to **find an aircraft design that satisfies performance constraints and completes the mission with feasible weight, fuel/energy, geometry**. It includes:

- Constraint Analysis
- Mission Analysis
- Weight Estimation
- Geometry Estimation
- Propulsion Sizing
- Iteration and Convergence

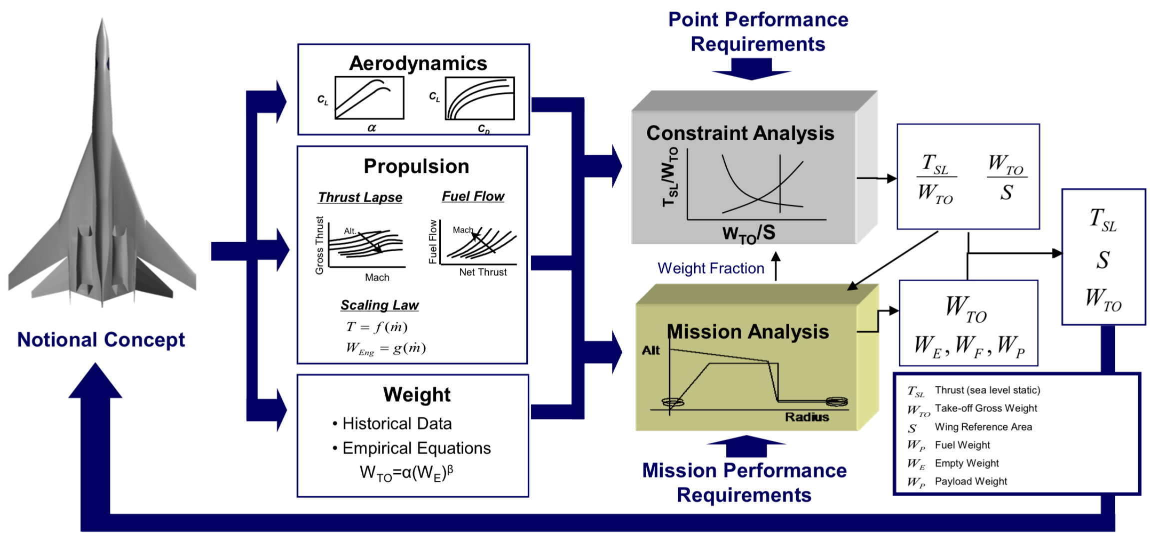

Figure 1: Sizing Process

2 Disciplines

2.1 Propulsion

In constriant analysis, we use lapse rate and TSFC, but more information needed. Take the jet engine as an example, the net thrust is defined as:

$$F_N = \dot{m}_a \left(v_j - v_0\right) \tag{1}$$

where F_N is the net thrust, $\dot{m}_a$ is the engine air mass flow rate, v_j is the jet exhaust velocity, and v_0 is the freestream velocity. The exhaust velocity v_j is obtained from the engine cycle analysis.

For our purpose, we use an existing engine and scale it up and down (**rubberized engine**) to meet our requirement (fixed cycle analysis). An scaling example is (A bigger fan produces more thrust by increasing $\dot{m}_a$):

$$F_N \propto \dot{m}_a \tag{2}$$

$$\dot{m}_a = \rho\pi\frac{d_f^2}{4}v_0 \tag{3}$$

Normally engine performance on various conditions are contained in an engine deck:

***h* (altitude)**	***M***	**% *Thrust***	***TSFC***

Figure 2: Engine Deck

To scale the engine, instead of using $\%Thrust$, we use $\frac{\%Thrust}{\dot{m}_a}$ to scale engine, which can be used to calculate the diameter of the fan. Also, instead of using TSFC, we use $\dot{m}_f$ for scaling.

Notice that some cases an existing engine is not suitable nor can easily be scaled, a new engine will need to be designed which is accomplished through an engine cycle analysis.

2.2 Structures

In conceptual design phase, this discipline is represented by calculating the total weigh and weight breakdown of the aircraft:

$$W_{TO} = W_E + W_P + W_F \tag{4}$$

$$W_E = W_{\text{wing}} + W_{\text{fuselage}} + W_{\text{empennage}} + W_{\text{prop}} + W_{\text{elec}} + W_{\text{hydraulic}} \tag{5}$$

Based on empirical study, we use the following relation to estimate the empty weight:

$$W_E = \alpha W_{TO}^{\beta} \tag{6}$$

where α and β are found for a given aircraft using historical data.

2.3 Aerodynamics

Typical aerodynamic characteristics of the airframe are needed:

- $C_{L_{max}}$
- Lift-drag polar
- $C_{L_{criuse}}$

Which normally from historical data.

3 Analysis

3.1 Constraint Analysis

Constraint analysis provides a solution mapped as a point in a thrust loading (T_{SL}/W_{TO}) agains wing loading (W_{TO}/S) diagram. It answers the question: What power/thrust and wing size do I need?

3.2 Mission Analysis

Mission analysis evaluates the aircraft over the mission profile, it answers the question: How much fuel or energy do I need?

4 Sizing Steps

4.1 Step 1

- It is assumed that W_P is known from the RFP.
- Payload weight may be comprised of passengers, crew, baggage, military loads, etc.

$$W_{TO} = W_E + \mathbf{W_P} + W_F \tag{7}$$

4.2 Step 2

Guess a value for W_{TO}:

$$\mathbf{W_{TO}} = W_E + \mathbf{W_P} + W_F \tag{8}$$

4.3 Step 3

Calculate S and T_{SL} using values of W_{TO} from Step 2, T_{SL}/W_{TO}, and W_{TO}/S from constraint analysis.

4.4 Step 4

Using empty weight regressions, find W_E/W_{TO} for guessed W_{TO}.

$$\frac{W_E}{W_{TO}} = f(W_{TO}) \tag{9}$$

$$\mathbf{W_{TO}} = \mathbf{W_E} + \mathbf{W_P} + W_F \tag{10}$$

4.5 Step 5

Calculate W_F available to fly the mission.

$$W_{F,\text{avail}} = \mathbf{W_{TO}} - \mathbf{W_E} - \mathbf{W_P} \tag{11}$$

4.6 Step 6: Fuel required calculation

- Split the mission into smaller segments and march through all segments of the mission.
- For any point in the mission, h and M are known.
- Using free body diagram, lift balances instantaneous weight. Therefore, wing area (S) could be obtained:

$$L = \frac{1}{2}\rho V^2 C_L S \tag{12}$$

- Now Drag (D) can be calculated:

$$D = \frac{1}{2}\rho V^2 C_D S \tag{13}$$

- Once drag at a point is known, part thrust can be calculated from the engine deck, and if it is linked to $\dot{m}_a$, the required mass flow rate of air can be obtained.
- The required fuel flow rate $\Delta\dot{W}_f$ for this segment can also be obtained from the table.
- Multiply this by the time of segment i to get the fuel burn for that segment, ΔW_{Fi}.

$$\text{Total fuel required} = W_{F,\text{req}} = \sum_i \Delta W_{Fi} \tag{14}$$

4.7 Step 7: Check convergence

- Check the difference between fuel available, $W_{F,\text{avail}}$, and fuel required, $W_{F,\text{req}}$.

$$\Delta W_F = W_{F,\text{avail}} - W_{F,\text{req}} \tag{15}$$

- If the difference is within tolerance, you have a converged solution. Now you have a sized vehicle.
- If not, update W_{TO} in Step 2 as:

$$W_{TO} = W_{TO} - \Delta F \tag{16}$$

or the weight difference can be split between $W_{F,\text{avail}}$ and W_{TO}

- In the calculations, it might turn out that there was not enough area.
  - You might want to update S as the largest area required or change the aerodynamics that can provide enough C_L.
- Update on the thrust would be $\dot{m}_a$ multiplier.
- Now, you have new values for T_{SL}, S, and W_{TO}.
- Repeat Step 4–Step 7 until $|\Delta F|$ converges to 0 or a specified tolerance.